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26 April 1971
SUPERSEDING
MIL-R-52183A
15 April 1963

MILITARY SPECIFICATION

RIPPERS, TRACTOR MOUNTING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers rippers for mounting on full-tracked tractors conforming to MIL-T-12229.

1.2 Classification. Rippers shall be of the following sizes as specified:

Size 4 - for size 4 tractors.
Size 5 - for size 5 tractors.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine (Heavy Duty).
MIL-T-3351	- Tractor, Full-Tracked, Low Speed; Tractor Wheeled, Agricultural; and Tractor Wheeled, Industrial: And Their Attachments, Packaging of.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).

FSC 3830

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MIL-L-10295	- Lubricating Oil, Internal-Combustion Engine, Sub-Zero.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-T-12229	- Tractors, Full-Tracked, Low-Speed, Diesel-Engine-Driven (Modified Commercial Design).
MIL-A-17076	- Angledozers and Bulldozers, Earthmoving: Cable and Hydraulic Operated, for Crawler Tractors.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-45199	- Lubricating Oil, Internal Combustion Engine (High Output Diesel).

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U. S. Military Property.
MIL-STD-209	- Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment.
MS39303	- Shank, Ripper, Rear Mounted: For Military Size 4 and 5 Tractors.
MS39304	- Tooth, Surface Ripping: For Ripper, Rear Mounted; for Military Size 4 and 5 Tractors.
MS39317	- Hose Assemblies; Hydraulic, Double Wire Braid Reinforcement, with Reusable Couplings.
MS39318	- Hose Assemblies; Hydraulic, 4-Spiral Wrap Reinforcement, With Reusable Couplings.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, N. Y. 10001.)

AMERICAN WELDING SOCIETY

B3.0 - Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX Welding Qualification.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

B117 - Salt Spray (Fog) Testing.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 Description. The ripper shall consist of mounting brackets, mounting hardware, a shank carrier beam, three straight shanks with replaceable teeth or teeth and points, and a hydraulic system (see 3.8). The shanks and teeth shall conform to MS39303 and MS39304 respectively. The rippers shall attach to full-tracked tractors conforming to MIL-T-12229, size 4 or 5, make, model, and series as specified. The ripper shall be maneuverable and controllable during operation.

3.2 First article.

3.2.1 Preproduction model. The supplier shall furnish a ripper for examination and test within the time frame specified, to prove prior to starting production, that his production methods and choice of design detail will produce rippers that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Materials and special treatments shall be utilized to prevent corrosion and to resist harmful effects of dust, fresh and salt water, and fungus.

3.3.1 High strength alloy steel. When high strength alloy steels are used, such alloys shall be identified in the technical manuals. Welding repair procedures for high strength alloy steels shall also be included in the technical manuals.

3.4 Ease of maintenance. All components subject to operator maintenance shall be accessible. Access shall be gained without removal of other components.

3.5 Structural adequacy. The ripper shall withstand the maximum drawbar pull of the tractor and the force of a pusher tractor of the same size on one shank without breakage or permanent deformation. The structural members and load carrying parts shall withstand all stresses imposed during operation as specified herein without breakage or permanent deformation.

3.6 Dimensions and weight.

3.6.1 Width. The overall width of the ripper shall not exceed the maximum width of the tractor by more than 2 inches and shall be not less than the track gage.

3.6.2 Height. When the ripper is mounted in the normal traveling position on the tractor, the ripper height shall not exceed that of the tractor.

3.6.3 Length. When the ripper is mounted on the tractor equipped with a dozer conforming to MIL-A-17076, type I, style B, size 4 or 5 as applicable, and in the travel position, the overall length of the tractor with dozer and ripper shall not exceed the length of the tractor and the dozer by more than 25 percent.

3.6.4 Weight. The overall weight of the ripper shall be not less than 5,000 pounds nor more than 12,000 pounds.

3.7 Ground clearance. When the ripper is mounted on the tractor and elevated to the normal travel position, the overall ground clearance specified in MIL-T-12229 for the applicable tractor shall be maintained.

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3.8 Hydraulic system. The hydraulic system shall include lines, fittings, cylinders and other hydraulic components required to provide the functions specified herein. The oil for the system shall conform to MIL-L-2104, grade 10 or MIL-L-45199, grade 10. For low temperature operation, oil conforming to MIL-L-10295 may be used. All component material shall be compatible with these oils and shall operate throughout an oil temperature range of minus 25° F. to plus 220° F. The temperature rise of the oil in any portion of the hydraulic system shall not exceed 100° F. at an ambient temperature of 50° F. and above. All system components shall withstand a minimum working pressure of 2,000 pounds per square inch (psi) and a minimum proof pressure of 4,000 psi without permanent deformation, damage, or external leakage and shall have a minimum burst pressure of 6,000 psi. Hydraulic component ports shall be sealed by the component manufacturer and shall remain sealed until final assembly of the ripper.

3.8.1 Cylinders. All hydraulic cylinders shall be double-acting, pin or socket mounted on self-alining bearings, or gimbal mounted, and shall be fabricated from seamless steel tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. Any cylinder used shall be protected against dust, rock, water, or any other material entering the cylinder. All internal cylinder ports shall be located beyond the area of piston travel. All cylinder port sizes shall limit the maximum oil velocity to 20 feet per second (fps) at maximum pump flow. Each cylinder shall have a rod wiper external to the rod seal to prevent foreign material and fluids from entering and damaging the seal and other cylinder components. The rod end cap on all cylinders shall be bolted to the cylinder. All cylinders shall have sufficient strength to perform as specified herein without buckling or bending. Internal flow restriction or by-pass shall be provided in the piston cylinder assembly when necessary to prevent damage due to shock loads. All bearing caps when used shall be bolted type with shims for wear take-up and shall be equipped with grease fittings. All gimbal mounts shall be equipped with replaceable bronze bushings and shall be equipped with grease fittings.

3.8.1.1 Cylinder tolerances. All hydraulic cylinder bores shall be machine honed or rolled within the limits of plus or minus 0.005 inch of the specified diameter. The cylinder inside diameter (ID) shall be round to within 0.003 inch. The surface roughness of the cylinder ID shall not exceed 32 microinches.

3.8.1.2 Cylinder performance. Each cylinder shall have a stroke of sufficient length to position the ripper as specified herein. Drift of each cylinder piston when supporting a load generating 2,000 psi minimum

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hydraulic pressure shall not exceed 8 inches per hour at an oil temperature of 150° F. (plus or minus 10° F.). Rod packing leakage may wet the rod but shall not collect on the rod packing retainer or rod end of the cylinder. Rod packing leakage shall not exceed 2.5 gram per inch of rod diameter per 1,000 cycles. The maximum pressure under no-load conditions required to actuate the piston shall be not more than 25 psi on the rod end and not more than 15 psi on the head end.

3.8.1.3 Piston and piston rod. Each piston shall have a phenolic (or similar material) bearing ring concentric with the piston to within 0.005 inch total indicator reading to carry the piston side loads against the cylinder walls and to prevent contact of the piston outside diameter (OD) and cylinder ID. The surface of the piston rod shall be hardened to 50 to 65 Rockwell C. The rod shall be hard chrome plated to resist corrosion. After plating, the maximum rod surface roughness shall not exceed 16 microinches. The plating shall be such that the rod will not corrode when exposed to the salt spray test in accordance with ASTM B117 for a period of 30 hours.

3.8.2 Hydraulic lines. Hydraulic lines shall consist of tubing, hose, and fittings. Hose shall be used where flexing and vibration may contribute to failure. All other lines shall be of tubing as specified herein. Hydraulic lines shall be secured to framing members by cushioned support clamps. The velocity of continuous flow oil in the pressure hoses shall not exceed 25 feet per second. Pressure hose and pressure hose assemblies in sizes smaller than 1 inch shall conform to MS39317. Pressure hose assemblies in sizes 1 inch and larger shall conform to MS39318. Pressure tubing shall be in accordance with SAE J524 or SAE J525.

3.9 Adjustments.

3.9.1 Depth adjustment. The size 4 ripper shall dig from ground level to a depth of at least 18 inches. The size 5 ripper shall dig from ground level to a depth of at least 24 inches.

3.9.2 Pitch variation. The ripper shall dig at a constant pitch, plus or minus 5 degrees, from ground level to maximum depth or shall have a minimum of two pitch adjustments.

3.10 Shanks. The ripper shall be equipped with three straight shanks having replaceable teeth or teeth and points and replaceable guards. When the shanks are the swiveling type, they shall be mounted to the carrier so that they will rotate a minimum of 10 degrees to the right and 10 degrees to the left of the center position.

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3.10.1 Shank mounting. One shank shall be mounted on the vertical centerline of the carrier beam, and the other two shanks shall be offset symmetrically from the vertical centerline.

3.11 Teeth or teeth and points. The teeth or teeth and points shall be replaceable without requiring special tools and equipment.

3.11.1 Fastener. A positive-locking pin shall be used to fasten the tooth or tooth and point on the shank. The pin shall be positioned so as to be protected from wear and abrasion. The pin shall prevent any movement of the tooth relative to the shank when the pin is installed. The pin shall withstand, without breakage or permanent deformation, the shear stress imposed by a lateral tensile load of 50,000 pounds applied to the tooth or tooth and point.

3.12 Push block. Unless otherwise specified, a push block, suitable for use with the pusher tractor equipped with a dozer conforming to MIL-A-17076, type I, style B, size 4 or 5 as applicable shall be furnished. The push block shall withstand the force imposed by the pusher tractor and transmit that force to the ripper. The push block shall be rigidly attached to the shank carrier beam.

3.13 Drawbar. When specified, a fixed drawbar in accordance with SAE J749 for industrial track-type tractor drawbar, size F, except for height, shall be furnished as an integral part of the ripper.

3.14 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.14.1 Lubricants. The ripper shall operate as specified herein when lubricated with military lubricants (see 6.5).

3.14.2 Lubrication fittings. Lubricating fittings shall conform to MIL-F-3541, types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand operated grease gun conforming to MIL-G-3859 with 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.14.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

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3.14.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a)	MIL-G-23827	- For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.
(b)	MIL-G-10924	- For bearings where operating temperatures do not exceed 175° F.

The ripper shall be assembled, tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.15 Lifting attachments. The ripper shall be provided with four permanently affixed lifting attachments. The attachments shall enable the ripper to be lifted in its normal traveling or operating position. The attachments shall be located so that a 1-inch clearance is maintained between the slings and all exterior parts of the ripper. Spreader bars are not permitted. Each attachment shall carry not less than 2-1/2 times the normal load imposed on the attachment. Attachments for multiple slings shall be located so that the slings will converge not more than 19 feet above the lowest extremity of the load. Attachments shall be positioned so that the lifting strain will be in line with the longitudinal axis of the eye of the attachment. Eye openings and clearance of the attachment eye shall conform to MIL-STD-209, type III. Attachments shall be beveled to prevent damage to the sling. Structural members to which the attachments are fastened and the fastenings shall withstand without permanent damage or permanent deformation, all stresses imposed by the attachment. Where required, structural members shall be reinforced to meet the lifting requirements specified herein.

3.16 Identification marking. The ripper shall be identified in accordance with MIL-STD-130. The marking shall be applied to the ripper on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

3.17 Instruction plates. Each ripper shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the ripper. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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3.18 Treatment and painting. The portions of the ripper and its components normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704. The final finish coat shall conform to color number 24087 as specified in FED. STD. No. 595.

3.19 Repair parts and maintenance tools. Such repair parts and maintenance tools as are specified shall be furnished.

3.20 Items to be furnished with the ripper. No shipment of rippers shall be made unless tools, repair parts, and accessories are included with the shipment or unless approval for shipment of the rippers without such items has been received from the contracting officer.

3.21 Government-loaned property. One of the following in the quantities indicated will be loaned by the Government (see 6.4):

Item No.	Description	Identification	Quantity
1	Tractor, full-tracked, low-speed, diesel-engine-driven, size 5, make and model as specified.	MIL-T-12229	2
2	Tractor, full-tracked, low-speed, diesel-engine-driven, size 4, make and model as specified.	MIL-T-12229	2

3.22 Workmanship.

3.22.1 Castings and forgings. All parts, components, and assemblies of the ripper which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.22.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.22.3 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.22.4 Welders and welding.

3.22.4.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.
Welding Qualifications of the ASME.

Suppliers who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, supplier's standard welder qualification may be substituted in lieu of the above codes provided that the supplier's procedure is equivalent to the above codes. The supplier shall be responsible for determining that automatic welder equipment operators are capable of producing quality welds.

3.22.4.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter other than mill scale that can be removed by chipping and wire brushing. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the rippers to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified, the welding process used in fabrication of the rippers shall be at the option of the supplier.

3.22.5 Machine work. Tolerances and gages for metal fits shall conform to the limits specified herein, and to the standards of the industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance

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of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance examination (see 4.4).
(c) Inspection of preparation for delivery (see 4.6)

4.3 Preproduction inspection.

4.3.1 Examination. The ripper shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The ripper shall be tested as specified in 4.5.2 using the two Government-loaned tractors (see 3.21). Failure of any test shall be cause for rejection.

4.4 Quality conformance examination. Each ripper shall be examined for the defect item No. 102 specified in 4.5.1. Evidence of a defect shall be cause for rejection.

4.5 Inspection schedule.

4.5.1 Examination. The rippers shall be examined for the following defects:

101. Dimensions and weight not as specified.
102. Parts or components missing or not as specified.
103. Parts or components do not function.
104. Safety devices or guards missing or not as specified.
105. Material not as specified.
106. Lubrication devices not as specified.
107. Treatment and painting not as specified.

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108. Accessories, identification, or other marking, operating instructions or manuals missing.
109. Workmanship not as specified.

4.5.2 Tests. Prior to testing, the ripper shall be limbered up as recommended by the supplier.

4.5.2.1 Tooth, point, and fastener. Subject one tooth or tooth and point and fastener to a lateral tension of 50,000 pounds. Pulling off of the tooth or tooth and point, or breakage or deformation of the tooth, point, or fastener shall constitute failure of this test.

4.5.2.2 Hydraulic cylinder. One cylinder shall be tested in accordance with test method No. 1 (see Appendix herein). The flow rate shall be equal to that produced at no-load governed engine speed at 2,000 psi. Evidence of any breakage or permanent deformation, excessive pressure required to move the rod end of the piston and the head of the piston, inability of the cylinders to operate as specified in 3.8.1, excessive wear of any component, evidence of corrosion, or evidence of contamination, galling, scuffing, or piston-to-cylinder contact as specified in test method No. 1 shall constitute failure of this test.

4.5.2.3 Structural. Attach the ripper to one of the Government-loaned tractors (see 3.21) and, using the other Government-loaned tractor as a pusher, strike an imbedded, anchored, concrete block or boulder with the center shank of the ripper at a speed of 3 miles per hour (mph). The block or boulder shall be of sufficient size to stall the two tractors. The minimum operating depth of the ripper for this test shall be 18 inches for size 4 and 24 inches for size 5. Repeat the procedure four more times. Repeat the procedure five times with the left end shank of the ripper, and then repeat the procedure five times with the right end shank. Repeat the procedure five times with the left and right end shanks simultaneously. Breakage or permanent deformation of the ripper, except tooth failure, shall constitute failure of this test.

4.5.2.4 Operation. Operate the ripper for a minimum of 50 hours as follows:

(a) 30 hours ripping rock having a minimum compressive strength of 10,000 psi at a minimum depth of 6 inches. The Government-loaned pusher tractor shall be used.
(b) 10 hours at maximum depth ripping in soil containing imbedded boulders.
(c) 10 hours ripping in compacted clay at maximum depth and at maximum tractor speed.

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Inability to maneuver or control the ripper, inability to perform the specified ripper operations, leakage of oil from the hydraulic system, evidence of breakage or permanent deformation, or nonconformance to 3.9 shall constitute failure of this test.

4.5.2.5 Lifting attachments. Apply the static load as specified in 3.15 for lifting attachments to each attachment when it is mounted on its structural member. Weld failure or permanent deformation of the attachment or the structural member shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking shall be examined and tested for compliance with MIL-T-3351 and section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Each ripper with components, repair parts and tools, shall be preserved, packaged, and packed in accordance with MIL-T-3351. The basic ripper shall be prepared as specified for attachments. The level of preservation and packaging shall be A or C and the level of packing shall be A, B, or C as specified.

5.2 Marking. Packages, shipping containers and unboxed rippers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The ripper is intended to be used as an attachment to a full-tracked tractor to dislodge boulders and other obstructions buried in the soil, for rock quarry operation, to tear up stretches of concrete and asphalt, and to loosen compacted soil.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size ripper required (see 1.2).
(c) Make, model, and series of tractor on which the ripper will be mounted (see 3.1).
(d) Time frame required for submission of preproduction model (see 3.2.1).
(e) When a push block is not required (see 3.12).
(f) When a drawbar is required (see 3.13).
(g) Applicable list of repair parts and maintenance tools (see 3.19).
(h) Level of preservation and packaging and level of packing required (see 5.1).

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6.3 Preproduction model. Any changes or deviations of production rippers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish rippers conforming to this specification.

6.4 Government-loaned equipment. The contracting officer should arrange to loan the property specified in 3.21.

6.5 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the ripper covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

Custodians:

Army - ME
Navy - YD

Preparing activity:

Army - ME

Project No. 3830-0086

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APPENDIX

FOR

RIPPERS, TRACTOR MOUNTING

Scope: The test method contained in this appendix is to determine whether rippers procured under this specification conform to the specified requirements.

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TEST METHOD NO. 1

HYDRAULIC CYLINDER COMPONENT TEST

1. TEST APPARATUS:

(a) Test bench that includes a means of automatically cycling each cylinder.
(b) 2,000 psi hydraulic power supply (same flow as pump installed on preproduction model).
(c) Pressure source, 4,000 psi minimum.
(d) Pressure gages, temperature gages, cycle counter, recorder.
(e) A load sufficient to develop a hydraulic pressure of 2,000 psi plus or minus 25 psi.
(f) A relief valve that will limit the full flow pressure to a maximum of 2,000 psi, and will reseat at a minimum pressure of 1,850 psi.
(g) Oil conforming to MIL-L-2104, grade 10.
(h) Filter, 10 to 25 micron nominal rating, full flow, with replaceable element.

2. TEST PROCEDURE:

(a) Proof pressure.
  (1) Position and mechanically hold the piston midway in the cylinder.
  (2) With the opposite port vented to atmosphere, apply an oil pressure of 4,000 psi to one side of the piston for 30 seconds. Check for external leakage.
  (3) Repeat (2) to the other side of the piston.
(b) Piston leakage.
  (1) With the cylinder filled with oil and the rod end port unrestricted, cap the head end port with a pressure pickup connected to the pressure recorder.
  (2) Mechanically load the head end of the piston to not less than 1,975 psi and hold this pressure for 15 minutes at an oil temperature of 150° F. plus or minus 5° F.
  (3) Measure and record the travel of the piston during the 15 minutes.
  (4) Repeat 2(b) (1), 2(b) (2), and 2(b) (3) for the rod end of cylinder.

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(c) Packing drag.
  (1) Fill the cylinder with oil at a temperature of 150° F. plus or minus 5° F.
  (2) Open the head end port of the piston to the atmosphere with the rod under no load except normal seal and wiper drag.
  (3) Pressurize the rod end of the cylinder and record the maximum pressure reached before the cylinder rod moves.
  (4) Repeat the above procedure by pressurizing the head end of the cylinder.

(d) Endurance.
  (1) Mount and support the cylinder on the test bench in a horizontal position or in its normal operating position. When two cylinders are used for a function in the system, and only one is being tested, the flow shall be reduced to 50 percent of that in 1(b).
  (2) Mechanically connect each cylinder to a load which will develop a pressure of 2,000 psi plus or minus 25 psi on each side of the piston. The oil temperature shall be 150° F. plus or minus 5° F. throughout the test. The minimum rate of pressure rise developed by the cycle valve shall be 20,000 psi per second. There shall be essentially no time delay at the end of each stroke, except when adjusting the stroke control mechanism.
  (3) Piston leakage. After completing endurance testing as specified in 2(d) (2), repeat the piston leakage test specified in 2(b). The ripper cylinder shall operate at full stroke at a rate of 15 cycles per minute for a total of 250,000 cycles.

(e) Disassembly and examination. Disassemble the cylinder and examine for contamination, evidence of galling, scuffing, and worn piston bearing ring permitting contact of the piston OD and cylinder ID.

(f) Subject all cylinder rods to the test specified in ASTM B117 for 30 hours. Examine the rods for corrosion.

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